Do wide binaries follow a scale set by the cosmic expansion? A registered protocol for Gaia DR4, with a dry run on DR3

Ugo Lanciano
Independent researcher, Locri, Italy · ORCID 0009-0008-7618-9635

Version 2.56, 2026-10-20. Archived on Zenodo, https://doi.org/10.5281/zenodo.23506401. Licence CC BY 4.0. Developed with the help of AI assistants, as declared in the record.

Student version. This PDF is the student version of the record, written with an AI assistant from the archived text. The archived text remains on Zenodo: https://doi.org/10.5281/zenodo.23506401

Abstract

This record registers, before Gaia DR4 (planned for 2 December 2026), a test of wide binaries. It is one of the checks of Mathematical Materiality (MM): the outcome concerns MM's reading of wide binaries, not the framework. Two MM readings are compared. The main one is a chain of correspondences that takes no parameter from the binaries. It links three things that seem unrelated: The field equation (QUMOND) is borrowed. The declared alternative is the reading of MM 2.54, which says that wide binaries are Newtonian. MOND with the external field is reported as a reference. The data and the rules are fixed now. The pairs of El-Badry, Rix and Heintz (2021) will be re-measured in DR4, with the cuts frozen on 11 October 2026. Stars with a DR4 non-single-star solution are removed. A forward simulation models hidden companions. The decision rule compares the chain with Newton and has written validity conditions: the test can favour one reading, or come out inconclusive or not valid.

Do wide binaries follow a scale set by the expansion of the universe? A registered protocol, with a test on today's data

Student version of MM record 2.56 (DOI 10.5281/zenodo.23506401), written with an AI assistant from the archived text; the archived text remains on Zenodo.

The archived text is in English.

Version 2.56 of the Mathematical Materiality (MM) series, October 2026. Mathematical Materiality, MM for short, is the author's framework of ideas. Each work in the series has a number.

What this record is. It is a *registration record*: it contains the protocol, the code, the data sample and a dry run.

- A *protocol* is the list of the steps of a test, written before doing it.
- *Registered* means written and fixed with its date before seeing the new data. In this way it cannot be changed afterwards.
- A *dry run* is a rehearsal on data already seen, to check that everything works.
- The record refers to MM 2.54 (DOI 10.5281/zenodo.23418156). 2.54 contains a *census*, that is, a list of the entries of MM. There, wide binaries are among the entries with a future check.

What is put to the test. This is one of MM's checks. The test can favour one of two MM readings. Or it can come out inconclusive, or not valid. Whatever the outcome, it concerns the reading of wide binaries, not the framework.

At the end you find how the work was done and a glossary.

Before starting: the words you need

Wide binaries. They are pairs of stars that orbit each other. The two stars are very far apart: thousands of times the

distance between the Earth and the Sun. The two stars attract each other very weakly. This is why they are a good test bench for gravity at very small accelerations.

Gaia. It is a satellite that measures the position and the motion of a great many stars. Its data come out in blocks, called *releases*. DR3 is the release already published.

Acceleration and the scale a_0. Acceleration says how much the velocity changes every second. It is measured in m s^{-2}, metres per second squared. In galaxies the stars far from the centre rotate too fast. Newton's gravity, with visible matter alone, predicts lower speeds. MOND is an idea of the physicist Milgrom. There is a very small acceleration, a_0 (read *a-zero*). Below a_0 gravity becomes stronger.

The boost. It is the number by which Newton's gravity is multiplied. The law is written ν(y) = 1/(1 − e^(−√y)) (ν is read *nu*). Here y is Newton's acceleration divided by a_0. When y is large, ν is almost 1 and Newton holds. When y is small, ν grows. The *asymptotic* boost of a reading, that is, the one of the widest pairs, is written η (read *eta*).

cH_0. c is the speed of light. H_0 (read *H-zero*) is the *Hubble constant*: it says how fast the universe is expanding today. The product cH_0 is an acceleration. Its value depends on the measurement of H_0: here 67.4 and 73.04 are used, in kilometres per second per megaparsec.

QUMOND and the external field. QUMOND is a version of MOND written with a *field equation*. The equation computes gravity at every point of space. A pair of stars near the Sun also feels the gravity of the Milky Way: it is the *external field*.

The words of the MM framework.

- A *reading* is an interpretation, written so that it can be put to the test.
- The *chain* is the main reading of this record: a row of choices linked like the links of a chain (section 1).

Likelihood. It says how probable a hypothesis makes the observed data. One uses its logarithm, the *log-likelihood*, written L. The higher it is, the better.

1. What is compared

The record compares six hypotheses. First the table, then the explanation.

hypothesis	content	role
chain, H_0 = 67.4	$a_0 = cH_0/6 = 1.091 \cdot 10^{-10}$ m s^{-2} (the two-mirror reading of MM 2.54); the reading-series law ν(y) = 1/(1 − e^(−√y)) (the MLS form of the radial acceleration relation); the Galactic external field from the rotation of the Milky Way (V = 233 km/s, R = 8.2 kpc: 1.966 a_0); the QUMOND field equation with all multipoles (borrowed, not derived)	**MM, main reading**
chain, H_0 = 73.04	the same with $a_0 = 1.183 \cdot 10^{-10}$ (1.814 a_0)	sensitivity
Newton	Newton's gravity	**MM, declared alternative** (the reading in MM 2.54)
MOND simple, MOND standard	QUMOND with the external field, $a_0 = 1.2 \cdot 10^{-10}$	reference
series, $a_0 = 1.2 \cdot 10^{-10}$	the reading-series law with the SPARC a_0 (earlier working reading)	reference

The chain with H_0 = 67.4: the main MM reading. It is made of four links.

1. $a_0 = cH_0/6 = 1.091 \cdot 10^{-10}$ m s^{-2}. It comes from the *two-mirror reading* of MM 2.54. 10^{-10} means divided by ten billion.
2. The boost law ν(y) = 1/(1 − e^(−√y)), called the *reading-series law*. It is the MLS form of the *radial acceleration relation*, the law observed in disc galaxies. MLS are the initials of McGaugh, Lelli and Schombert.
3. The external field of the Galaxy, derived from the rotation of the Milky Way: V = 233 km/s is the rotation speed of the Galaxy at the distance of the Sun. R = 8.2 kpc is its distance from the centre. The field is worth 1.966 a_0.

A kpc (kiloparsec) is a thousand parsecs; a parsec is a little more than three light years.

4. The QUMOND field equation, with all the *multipoles*. The multipoles are all the details of the shape of the field. It is borrowed from MOND, not derived from MM.

The other hypotheses.

- **The chain with $H_0 = 73.04$.** It is the same, with $a_0 = 1.183 \cdot 10^{-10}$. The external field is then worth 1.814 a_0. It serves to see how much the result depends on H_0: it is a *sensitivity* test.
- **Newton.** Newton's gravity, without boost. It is MM's *declared alternative*: it is the reading that MM 2.54 gives for binaries.
- **MOND simple and MOND standard.** QUMOND with the external field and $a_0 = 1.2 \cdot 10^{-10}$. They are two ways of writing the passage between Newton and MOND. They serve as reference.
- **The series, with $a_0 = 1.2 \cdot 10^{-10}$.** It is the reading-series law, with the a_0 measured by SPARC. SPARC is a catalogue of galaxies. It was an earlier working reading. It serves as reference.

What is special about the chain. It takes no parameter from the binaries. It links three things that seem unrelated:

- the expansion of the universe, through $a_0 = cH_0/6$;
- the law read in galaxies;
- the rotation of the Milky Way.

The expected boost. The *asymptotic* boost of the chain is the one of the widest pairs. It is $\eta = 1.327$ with $H_0 = 67.4$. It is 1.362 with $H_0 = 73.04$. In words: for the widest pairs gravity would be 1.327 times Newton's, with $H_0 = 67.4$.

2. The data

The pairs.

- They are the wide binaries of the catalogue of El-Badry, Rix and Heintz (2021).
- They were downloaded on 16 October 2026, with three conditions: $\varpi_1 > 4$ mas, sepAU > 200, R < 0.1. There are 167,301 pairs.
- ϖ (read *variant pi*) is the *parallax*, the small apparent shift that gives the distance; the mas is a thousandth of an arcsecond, a very small angle;
- sepAU is the separation between the two stars in astronomical units;
- R is the probability that the two stars are close only by chance.
- With the frozen cuts, the sample reproduces pair by pair the one used during development: 913 low-acceleration pairs and 21,489 control pairs.

The values of the new release. The protocol was deposited before Gaia DR4. It is the next release of Gaia data, planned for 2 December 2026. The same pairs will be re-measured in DR4.

- The DR3→DR4 *neighbourhood table* is used. It says which star of DR4 corresponds to which star of DR3.
- If that table is missing, the stars are searched for by position. The epoch J2017.5 is used, with a maximum distance of 0.3″ (0.3 arcseconds).
- Only *one-to-one* links are kept: one star of DR3 with only one star of DR4.
- The stars that in DR4 have a *non-single-star* solution are removed: Gaia does not describe them as simple, isolated stars. This is a new selection, not simulated.

The conversion. The DR4 values are put into the format used during development. Then they go through the same chain of programs, with the *quality cuts* frozen on 11 October 2026:

- BinType = MSMS: both stars are on the *main sequence*, that is, stars that burn hydrogen in their centre, like the Sun;
- d < 200 pc: the distance from us is below 200 parsecs;
- ϖ/σ_ϖ > 20: the parallax is measured very well;
- parallaxes of the two stars consistent within 3σ (σ is the typical error);
- 3 < M_G < 12: M_G says how luminous the star is;
- error on u < 0.1. Here u is the relative velocity of the two stars, seen on the sky, divided by the velocity of a Newtonian circular orbit at their separation seen on the sky;
- depth term < 0.1;
- u < 3.

3. How the comparison is made and how the decision is taken

The exact rules are in the protocol. Here we summarise them.

The forward simulation. One simulates on the computer what should be seen with each hypothesis.

- The *hidden companions* are taken into account: third stars nearby that are not seen but change the measurements. These are simulated:
- the fit of the position;
- the quality index RUWE;
- El-Badry's cut;
- the 2+2 systems, that is, two pairs bound to each other;
- the *near-twin* pairs, of two almost equal stars.
- There are two *bands*. The control band has 4000 random pairs with $x \geq 10$. The low-acceleration band has $x < 1$. Here x is the Newtonian acceleration of the pair divided by a_0.
- There is a *grid* of cases. The twins are 4 % or 20 %. The companions per star go from 0 to 0.4 (0-0.4). The *replicas* are 3: three repetitions with different random numbers.
- In all: 6 hypotheses, 60 files.

The likelihood. Poisson statistics is used on 26 classes per band. Each class has the same number of pairs.

- The twin fraction is common to the two bands. The *mass scale* is also common: it goes from 0.85 to 1.15.
- The companion fraction is separate for each band.

Validity. The test is not valid if one of these conditions is missing.

- **V1:** at least 600 pairs in the low band and 15,000 in the control band.
- **V2:** the *libraries* must have 200,000 computed orbits. The error on the energy must be below $2 \cdot 10^{-2}$ at the 99th percentile, that is, in 99 per cent of the orbits. No pair must start unbound. The grid must be complete. The fingerprints of the data must be equal.
- **V4:** the best hypothesis must have a small *deviance*. The deviance measures how far a hypothesis departs from the data. In each band it must be at most 3 times the number of classes ($\leq 3 \times$).

The decision. D = L(chain 67.4) − L(Newton): the difference in log-likelihood between the chain and Newton. D_k is the same difference, redone removing one replica.

- **Chain favoured:** $D \geq +5$ and every $D_k \geq +2$.
- **Newton favoured:** $D \leq -5$ and every $D_k \leq -2$.
- **Otherwise:** inconclusive.
- E = L(chain) − L(best MOND) is also reported. The chain is "close to the best MOND" if $E \geq -5$.

4. The dry run on today's data (DR3)

These data had already been seen during development.

Same data, same simulations.

- The DR3 values were written in the format of the Gaia archive and then converted. They give the same pairs and the same quantities as the development files: the difference is 0.
- The simulations for Newton and for the chain were redone, with the same row order. They are identical to the ones that already existed.

The rules applied to the 60 existing DR3 simulations.

hypothesis	ΔlogL from the best	deviance control / low
MOND simple	0	43 / 21
MOND standard	−0.8	28 / 38
chain, $H_0 = 67.4$	**−2.4**	38 / 31

chain, $H_0 = 73.04$	−4.2	45 / 28
series, $a_0 = 1.2 \cdot 10^{-10}$	−8.7	49 / 33
Newton	**−10.0**	37 / 47

Let us read the table. ΔlogL is the difference in log-likelihood from the best hypothesis: 0 for the best, negative numbers for the others.

- **MOND simple** is the best: 0. Deviance 43 in the control band and 21 in the low band.
- **MOND standard:** −0.8 (28 / 38).
- **The chain with $H_0 = 67.4$:** −2.4 (38 / 31).
- **The chain with $H_0 = 73.04$:** −4.2 (45 / 28).
- **The series with $a_0 = 1.2 \cdot 10^{-10}$:** −8.7 (49 / 33).
- **Newton:** −10.0 (37 / 47).

The numbers of the rule.

- The test is valid.
- D = +7.64, with the subsets +7.56, +4.50 and +2.07.
- E = −2.36: the chain is close to the best MOND.

What the rule would say on DR3. The decision would be "chain favoured over Newton". But the minimum margin is only just met: the lowest subset, +2.07, exceeds the threshold +2 by a little.

The download. The passage from DR3 to DR4 was tried on DR3, both by identifier and by position.

What could not be tested before DR4.

- The DR4 neighbourhood table itself.
- The effect of the selection of non-single stars.
- A full rerun of the grid on the new data.

5. Grade and declared limits

Grade B*. The DR3 data were partly seen before registration. And DR4 reuses DR3 stars and observations. So a favourable outcome will be a confirmation on non-independent data, not a blind prediction.

The limits.

- The *generator* of the simulated pairs was not rewritten from scratch. The *tails* at 3σ, that is, the most extreme cases, were not studied.
- The masses have a free common scale, not full distributions.
- The *eccentricity*, that is, how flattened the orbits are, is fixed according to Hwang et al. 2022.
- QUMOND is borrowed.
- The Galactic V and R are fixed.
- The selection of non-single stars is not simulated.
- The correlation between the two proper motions, pmRA and pmDE, is not used. They are the motion on the sky in the two directions.

The scope of the test was reduced on purpose: this is one check among many.

The amendments. Any change after the deposit will be a dated addendum. It will have its reason and the outcome under both rules: the original one and the new one.

How the work was done

This section gathers the procedure. In the archived text it corresponds to sections 6 and 7 and to the file names of

sections 2 and 4.

Who did the calculations. The author is not a professional physicist. The calculations were done with AI assistants (Claude, with two checks by Codex). No human *referee* reviewed them. A referee is an expert who checks a work before publication.

The verifications: two rounds of Codex/GPT-6, another AI assistant. The reports are in documenti/verifica_astra/.

- The first round, on version 1 of the protocol, gave "not ready", with nine points. They were all addressed in version 2. The list is at the end of the protocol.
- The second round, on version 2: eight points resolved, no new error. One point was partly open: an explicit check of unbound Newtonian orbits. And a clean "not valid" on an incomplete grid, instead of a crash of the program.
- After the second round (valuta_dr4.py 2.1):
- the Newtonian library is checked orbit by orbit: 0 unbound states out of 200,000;
- an incomplete grid now gives "not valid". This was tested with a missing file and with a file of two replicas (risultati/prova_a_secco/prova_valutazione_casi.log).
- There was no third round.

The registrations. Every file was registered before being run. The registration has a SHA-256 fingerprint and the UTC time (documenti/registri/). A SHA-256 *fingerprint* is a code computed from the content of a file: if the file changes even slightly, the fingerprint changes.

The reference text. The protocol is documenti/PROTOCOLLO_DR4_v2.md, in Italian. The archived text, in English, summarises it.

The files cited in the sections.

- Section 2: the query and the SHA-256 of the download are in dati/. The selected sample, with the Gaia identifiers, is dati/campione_F.csv. El-Badry's catalogue is on VizieR, an archive of astronomical catalogues: J/MNRAS/506/2269. The download of the DR4 values is codice/registrazione/scarica_dr4.py; the conversion is converti_dr4.py, which puts the data in canonical order and with the SHA-256 fingerprints.
- Section 4: the download tests are in risultati/scarico/.

What is in the package.

- `codice/registrazione/`: the download, conversion, simulation, evaluation and dry-run programs (version 2; version 1 is in versione_1/).
- `codice/pipeline/`: the programs of the development chain that they call, unchanged. The code contains the author's paths (E:...), to be adapted.
- `dati/`: the query, the SHA-256 of the download, the selected sample with the Gaia identifiers.
- `risultati/`:
- the libraries: dynamic checks and SHA-256. The six libraries (217 MB each) are not included. They can be regenerated with orbite.py or orbite_mm.py. The *seeds* are 1, 12, 13, 14, 15, 16. FATT_PASSI is 1 for Newton and 4 for the others. The seed is the starting number of the random numbers;
- the sample, the chain, the dry run, the download tests.
- `documenti/`: the plan, the decision on the main reading, the protocol versions, the verification report, the registers.

The licences. Texts and materials: CC BY 4.0. Code: MIT.

This version. This student version was written with an AI assistant (Claude) from the archived text in English. First a full Italian translation was made. Then the translation was rewritten for a school student. The numbers were checked with a program: the translation against the English text, this version against the translation. The archived text is the reference.

This English text is a translation of the Italian student version, which the author read and approved.

Glossary

The words of the framework (MM)

- **Chain.** The main reading of this record: $a_0 = cH_0/6$, the reading-series law, the field of the Milky Way and QUMOND.
- **Reading.** An interpretation, written so that it can be put to the test.
- **Registered.** Written and fixed with its date before seeing the new data.

The words of physics

- **a_0.** A very small acceleration: below a_0, in MOND, gravity is stronger than Newton's. In MM $a_0 = cH_0/6$.
- **Acceleration.** How much the velocity changes every second.
- **Boost (ν, η).** The number by which Newton's gravity is multiplied; η is its asymptotic value, for the widest pairs.
- **Deviance.** A number that measures how far a hypothesis departs from the data.
- **Dry run.** A rehearsal on data already seen, to check that everything works.
- **External field.** The gravity that comes from outside a system; for binaries near the Sun it is that of the Milky Way.
- **Gaia, DR3, DR4.** Gaia is a satellite that measures positions and motions of stars; DR3 is the data release already published, DR4 the next one.
- **Hidden companions.** Stars close to one of the two stars of the pair, which are not seen but change the measurements.
- **Log-likelihood (L).** The logarithm of the likelihood: the higher it is, the better a hypothesis accounts for the data.
- **MOND, QUMOND.** MOND is Milgrom's idea: below a_0 gravity is stronger. QUMOND is a version of it written with a field equation.
- **Parallax (ϖ).** The small apparent shift of a star, which gives its distance.
- **Protocol.** The list of the steps of a test, written before doing it.
- **Replicas.** Repetitions of a simulation with different random numbers.
- **Sigma (σ).** The typical error of a measurement.
- **Wide binaries.** Pairs of stars very far apart that orbit each other.

References

Works cited in the record, with their DOI.

- Banik et al. (2018). Testing gravity with wide binary stars like α Centauri. https://doi.org/10.1093/mnras/sty2007
- Banik et al. (2023). Strong constraints on the gravitational law from Gaia DR3 wide binaries. https://doi.org/10.1093/mnras/stad3393
- El-Badry et al. (2021). A million binaries from Gaia eDR3: sample selection and validation of Gaia parallax uncertainties. https://doi.org/10.1093/mnras/stab323
- Hwang et al. (2022). The eccentricity distribution of wide binaries and their individual measurements. https://doi.org/10.1093/mnras/stac675
- Mardling et al. (2001). Tidal interactions in star cluster simulations. https://doi.org/10.1046/j.1365-8711.2001.03974.x
- McGaugh et al. (2016). Radial Acceleration Relation in Rotationally Supported Galaxies. https://doi.org/10.1103/physrevlett.117.201101
- Mikkola et al. (2002). A Time-Transformed Leapfrog Scheme. https://doi.org/10.1023/a:1021149313347
- Milgrom (2010). Quasi-linear formulation of MOND. https://doi.org/10.48550/arxiv.0911.5464
- Pecaut et al. (2013). INTRINSIC COLORS, TEMPERATURES, AND BOLOMETRIC CORRECTIONS OF PRE-MAIN-SEQUENCE STARS. https://doi.org/10.1088/0067-0049/208/1/9
- Raghavan et al. (2010). A SURVEY OF STELLAR FAMILIES: MULTIPLICITY OF SOLAR-TYPE STARS. https://doi.org/10.1088/0067-0049/190/1/1
- Vallenari et al. (2022). Gaia Data Release 3. https://doi.org/10.1051/0004-6361/202243940